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Historical roots of the psychological laboratory

The birth date of modern psychology is usually placed toward the end of 1879 when Wilhelm Wundt designated some space at the University of Leipzig to be used for the conduct of psychological experiments.¹ Of course, the date is arbitrary, as are all such birth dates for disciplines. This arbitrariness arises less from the rival claims of other locations or individuals than from the obvious fact that the birth of a discipline is not a singular event but a complex process extending over a considerable period of time. In the case of psychology the relevant period extends both before and after the magic date.

When psychology became an autonomous field of research it did not invent its concepts and problems out of the blue but took them over from already existing fields like philosophy and physiology. Similarly, the practical activities that came to be identified as methods of psychological research were anything but completely new inventions. They were more in the nature of adaptations of already existing practices to a somewhat different context. Those who laid the foundations of a new psychological research tradition were constrained at every step by the traditions of investigative practice with which their general cultural experience or their personal training had made them familiar. The best they could do was to modify these practices, sometimes in quite minor ways, to suit the new goals they had in mind.

When Wundt proceeded to give practical shape to his program for a new kind of psychology, he based himself on three traditions of investigative practice, two of them explicitly, the other implicitly. Explicitly, he based himself first on the specific form of experimental practice that had recently grown up in physiology. This provided him with certain material techniques and a certain way of asking research questions. Second, he proposed to apply these procedures to a different object of investigation than the objects

addressed by physiological experimentation. The object he proposed was the private individual consciousness. Historically the notion that this could be a special object of investigation was closely linked with the investigative practice of introspection. Wundt rejected introspection in its traditional form, but he did accept its object. Hence in the present chapter we will examine the historical roots of the notion that the “inner” world of private experience could be methodically explored. Wundt’s own adaptation of the introspective method will be discussed in chapter 3, in the context of the controversy that erupted around this topic a generation after the establishment of his laboratory.

Although it received far less discursive attention than introspection or experiment, a third element in Wundt’s investigative practice was absolutely crucial to the whole enterprise – the social organization of psychological experimentation. This aspect provides the strongest grounds for locating the beginnings of experimental psychology in Wundt’s laboratory, for it was here that scientific psychology was first practiced as the organized and self-conscious activity of a community of investigators. Moreover, this community spawned others similar to itself, thus initiating a specific tradition of investigative practice. This development, more than anything else, established modern psychology as a distinct field of inquiry that was the preserve of an identified research community. Wundt established the first community of experimental psychologists simply by adopting the prevailing German university link between teaching and research in the form of an institute where students could pursue their research. This resulted in the emergence of certain social patterns of psychological experimentation, which quickly became traditional. That development will be examined in the final section of this chapter.

Introspection and its object before Wundt

Although introspection has not played a large role in the development of psychology during most of the twentieth century, it was a very important topic of methodological discussion at the time of the emergence of modern psychology and for several decades thereafter. Far from being the time-hallowed method of prescientific psychology, as it was often presented by behaviorist crusaders, introspection is actually a historically recent invention. Its roots may go back to certain tendencies in Protestant theology that advocated a methodical self-examination of one’s conscience. (Such a practice would take over some of the functions of the public confessional.) In English the term does not appear to have been used to refer to a systematic and ethically neutral practice of self-observation until well into the nineteenth century, and it certainly was not in general use until the second half of that century.

Of course, philosophers before that time appealed to the subjective self-

awareness of their readers, just as they appealed to their readers' observation of the actions of others, to their sense of logic, and to their religious and ethical beliefs. But these appeals were not regarded by them, nor can they be regarded by us, as constituting a methodology.² Philosophers' appeals to common experiences of subjective self-awareness were no more based on any specific methodology of introspection than their appeals to common experiences of social life were based on any specifically sociological methodology.

One must distinguish between our common everyday fund of experiences and those deliberate and systematic arrangements for the methodical production of certain kinds of experiences that we call methodologies. In the latter sense there certainly was no introspection before the nineteenth century.³ The lexical change already noted appears to have reflected a change in the thing referred to, a new awareness of the methodological aspects of self-observation. In the English-language literature that awareness appears to have come about in response to certain Continental criticisms of the traditional philosophy of mind,⁴ which went back to John Locke's distinction between two sources of knowledge – sensation and reflection. The former gave us knowledge of the external world, whereas the latter gave us knowledge of the operation of our own minds. Thus, to the philosophy of nature, based on the evidence of our senses, there corresponded a philosophy of mind based on the evidence of reflection. But this philosophical tradition made no distinction in principle between mere awareness of mental states and deliberate observation of such states.

That distinction was the product of Immanuel Kant's attempt at constructing a philosophical synthesis that would transcend the conflict of empiricism and rationalism. He accepted from Lockean empiricism the notion of a world of private experience, which becomes manifest to its possessor through the medium of an "inner sense," analogous to the outer senses that give us experience of the external world. But then he raised a question Locke had not raised: Can the experiences conveyed by the inner sense form the basis of a mental science just like the experiences of the outer senses form the basis of physical science? The answer was a decided no, because science, unlike everyday experience, involves a systematic ordering of sensory information in terms of a synthesis expressed in mathematical terms. The material provided by the inner sense was, however, resistant to mathematization, and so there could not be a science of mental life, or psychology.⁵ Thus, although Kant did not deny an empirical basis to psychology, he considered that this was not enough to establish it as a science. For that a particular methodological element would have to be added, and in the case of mental life this addition would not work.

In addition to making a fundamental distinction between natural science and psychology, Kant made another, more important distinction for the fate of psychology in the nineteenth century – that between the very dif-

ferent domains of psychology and philosophy. In the Lockean tradition of mental philosophy that distinction did not exist, and therefore psychology lacked any clear identity as a special field of study with its own objects and goals. The Lockeans were primarily interested in providing an empiricist theory of knowledge, and in doing so they appealed to a particular theory of mental functioning. But they did not distinguish between the psychological problem of how particular mental contents were caused in the natural world and the logical or philosophical problem of how such mental contents could be said to constitute knowledge of the world.⁶ As long as that distinction was not made, psychology and philosophy remained indissolubly fused and the question of a specifically psychological methodology could not arise. Empiricist philosophers could go on appealing to their readers' everyday self-awareness as they appealed to other aspects of their experience, without any feeling that this required any special methodology.

Kant, however, made a sharp distinction between mental life as it is present to subjective self-awareness and the general principles in terms of which that life is organized. The former is part of the empirical world, just like our perceptions of external reality, but the latter points beyond such an empirical world to a "transcendental ego" that is the source of the fundamental categories that characterize human experience in general. To use a somewhat simplified illustration, there is a huge difference between examining the factors involved in particular spatial perceptions and examining the implications of the fact that our perceptions are characterized by spatiality. With this distinction Kant clearly separated the domains of philosophy and psychology and thereby raised the question of psychology as a nonphilosophical empirical discipline.⁷ Such a discipline would be empirical, because it would base itself on the evidence of the "inner sense" in the same way that the various natural sciences based themselves on the evidence of the external senses.

But Kant did not think that an empirical basis was sufficient to establish a field as a true science. The classificatory botany of his day, for example, was certainly empirical but Kant did not regard it as a real science. It simply collected specimens and categorized them in terms of the occurrence and cooccurrence of particular features. But it was unable to show why the features of the plant world took on the arrangement they did rather than some other equally possible arrangement. The Newtonian science of classical mechanics, on the other hand, was able to demonstrate mathematically why a system of moving physical bodies like the planets maintained a particular arrangement of its parts, given certain initial assumptions. What made mechanics a true science was the mathematization of its subject matter. Kant had reason to believe that the data of the "internal sense" would always resist mathematization, and so it followed that psychology would never be a true science.

Kant's contribution to the history of psychology was therefore ambiguous. Although he provided the field with a clear identity, separate from philosophy, he also awarded this field a rather low status. At best, it would be little more than a collection of contingent rules, a kind of natural history of the mind, lacking both the fundamental importance of philosophy and the rational consistency of science.⁸ To become a science, its special method of introspection would have to yield to mathematical treatment in the way that the visual data of astronomy, for example, yielded to mathematical treatment. But this would not happen, and so the subject had no future as a science.

The nature of the change from Locke to Kant is worth examining a little more closely. Clearly, the change involved a set of interconnected distinctions, a process of differentiation in a certain sphere of intellectual work. Kant's writings expressed a new level of interest in the organization of intellectual work. Not only was there the usual modern interest in the knowledge acquisition process as such, but there was a recognition of the fact that there are different kinds of knowledge based on different practices. It was now accepted that intellectual inquiry divided into different fields, each of which was constituted by a trio of interdependent factors: a certain object of investigation, a particular type of question or problem characteristic of the field, and a specific methodology by means of which these questions are pursued. Although the definition of the object of investigation limits the problems that can be formulated with respect to it and suggests certain methods as appropriate for their investigation, the adoption of these methods will continually re-create the objects that they presuppose and constrain the questions that can be legitimately asked of them. Thus, the advance of introspection from an occasional use of common experiences to a self-consciously employed *method* of investigation would not have been possible without the previous definition of a part of experience as *inner* experience, but of course the systematic practice of introspection helps to create just such a world.

It is sometimes said that modern psychology owes its origin to the post-Renaissance dichotomy between the extended mechanical world of matter and the totally different world of the mind,⁹ but this is only a half-truth. That dichotomy, which is also the point of departure for the Lockean philosophy, certainly created a potential object of specialized investigation, the inner world of the isolated individual mind. But in the philosophy of Descartes and Locke it remained a *potential* object of technical scrutiny. What these philosophies did was to propose a certain way of talking about human experience, a system of terms implying certain distinctions and divisions in our experience of life.¹⁰ This was an indispensable first step in the construction of a new field of study, but it was only a first step. Within a language community that tagged certain experiences as "inner" and treated them as fundamentally distinct, the potential for a specialized study

of such experiences existed. But for this potential to be translated into actuality a further development was necessary.

This development involved the elaboration of a certain organization of intellectual work that went beyond the creation of new language communities. It was an organization manifested in the growth of the sciences during the eighteenth century. As knowledge of the natural world accumulated, scientific inquiry gradually became more differentiated and more self-conscious. The question of the relationship between science, or natural philosophy, and the more traditional concerns of philosophy began to be of interest, particularly in Germany where scientific activity was relatively more likely to occupy members of university philosophy faculties. The result was a sharpened emphasis on the methodological element in the constitution of the various fields of study. Among other things Kant was clearly concerned with the professional role of the philosopher in an intellectual situation that was being rapidly changed by the prestige and the spread of natural knowledge. A major effect of some of his writings was to give to philosophy a powerful role in the adjudication of divergent knowledge claims.¹¹ With Kant there began a tradition that was to last for well over a century, where the academic philosopher assigned relative status and intellectual limits to the various fields of empirical study. The grounds on which he did this were largely methodological, and this produced a heightened concern with such matters among those who were forced to justify themselves before this court.

The result was that in the nineteenth-century German universities the question of psychology as a field of study became problematized in methodological terms. Three issues dominated this debate: introspection, mathematization, and experimentation. The first two went back to Kant, the third became important after the middle of the nineteenth century. Friedrich Herbart, Kant's successor at the University of Königsberg, engaged in a monumental exercise to show that the mathematization of psychology was possible in principle,¹² and Herbartian psychology remained an influential school until the end of the century.

Introspection remained a contentious issue. It will be remembered that for Kant the examination of the individual consciousness produced only limited results in both a scientific and a philosophical context. Its philosophical limitations arose out of the limitations of all empirical knowledge. The evidence of the "inner sense" was just as much subject to these limitations as the evidence of the external senses. In both cases the knowledge conveyed was knowledge of phenomena, of appearances, while the real world remained hidden from view. Introspection was limited to the phenomenal self: "I know myself by inner experience only as I *appear* to myself."¹³ The true basis of our mental life, however, the subject of pure apperception, cannot be examined by relying on "inner experience."

The influence of the Hegelian school in post-Napoleonic Germany only

served to confirm the low evaluation of individual self-observation. With *Geist* (mind or spirit) conceived as an objective category embodied in cultural formations and manifesting itself in philosophical abstractions, the role left to the observation of the individual human mind was small indeed.

In general, nineteenth-century philosophers divided into two camps on the subject of introspection, depending on which aspect of Kant's ambiguous legacy they chose to emphasize. Some, like Hegel and Kant himself, devalued the philosophical significance of the evidence from inner experience, others were encouraged by the possibility of developing a new empirical discipline on this basis.¹⁴ The critical factor in this divergence of viewpoints was always the degree of individualism that characterized each philosophical position. According to one tradition, represented in its purest form by the British empiricists, philosophical speculation had to be grounded in the experience of individual minds. According to another tradition, represented by German Idealism, philosophy was grounded in supraindividual principles of order.¹⁵ The latter group rejected introspection, blaming the superficiality and unreliability of its results; the former group accepted introspection as a necessary basis of philosophy and psychology. The second group made a clear division between philosophy and psychology, whereas the empiricists tended to blur the distinction between psychological and philosophical issues. Where the self-conscious individual was regarded as the centerpiece around which the world had to be arranged, introspection was held in high regard as a method;¹⁶ where the order of the world took precedence over the individual, the examination of the individual consciousness was regarded with suspicion or condescension.

As the order of the world always included the social order as an important, and often as the all-important, component, conceptions of the role of introspection were not unconnected with considerations of political philosophy. During most of the nineteenth century a positive attitude to introspection tended to go with a philosophy of liberal individualism, while negative attitudes were more likely to be found among those who stressed the priority of collective interests or institutional requirements.¹⁷

Even an apparently private technique like introspection turns out to be very much a form of social action. It achieves this on two levels. On an immediate, more technical level introspection becomes social performance insofar as its use as a methodology requires the employment of language for the categorization and communication of experience.¹⁸ But here we are not directly concerned with this inescapable fact but with the consequences of introspection's historical status as an avowed method of collective inquiry leading to true knowledge. To discuss introspection as a *method* means discussing it as a *collective* project, and that means putting it in the context of cognitive goals and interests that are assumed to have a shared, social character. It is not introspection as an idiosyncratic habit that is of interest here, but introspection as a social project. As such, it occurs in particular

historical contexts from which it derives its meaning. This is true even when it is practiced unreflectively, as long as there is a significant uniformity of practice and of context. This context endows introspection with a variety of social meanings and hence also determines the forms of actual introspective practice.

As we have seen, the emergence of the notion of introspection as a *method* was intimately linked to the emergence of psychology as a separate field of study with its own special subject matter. The belief in the existence of this subject matter, the private world of inner experience, was a precondition for any meaningful discussion of introspection as a method. And such discussions, as well as the actual practice of the method, tended to validate this belief. Thus, method and object of investigation mutually confirmed each other, a state of affairs we will meet again in connection with other methods. Different evaluations of the object led to different evaluations of the method. So there could be differences of opinion, not only about whether introspection was valuable but about what one was actually doing when engaging in this practice. Thus, introspection could take on different meanings, depending on how its object was seen as fitting into the scheme of things. But this in turn depended on general or specific social interests. In this respect too introspection was no different from other forms of social practice to which the status of “method” was assigned.

Experimental physiology and the investigation of functions

Discussions of experimentation in psychology have often suffered from a misplaced abstractness. Not uncommonly, they have referred to something called *the* experimental method, as though there was only one, and in the worst cases this singular category is identified with another, called *the* scientific method. This kind of abstraction can only be maintained by ignoring the history of science, which provides evidence for the existence of several kinds of experimentation.¹⁹ Moreover, the different kinds of experimentation have quite varied historical roots. Therefore, an inquiry into the origins of psychological experimentation must be concerned with the specific historical forms of experimentation that played a role in the constitution of the new field. Although experiments in the physics of sound and optics undoubtedly affected early laboratory practice in psychology, the systematic and programmatic aspect of psychological experimentation was clearly not derived from physics but from physiology. Indeed, in the first exemplar of a psychology constructed on an experimental basis, the term “physiological psychology” functioned as a synonym for experimental psychology.²⁰ Wilhelm Wundt frequently talked about adopting the experimental method from physiology. Let us see what was being imported into psychology here.

When Wundt received his medical training, just after the middle of the

nineteenth century, the experimental method had only barely established itself as the primary method of physiological investigation in Germany.²¹ The older generation of physiologists represented by Johannes Müller (d. 1858) did not regard physiology as an *essentially* experimental science, although they might perform experiments from time to time. It was Müller's ablest students and their generation who changed that. But these men – Du Bois-Reymond, Ludwig, Helmholtz – were only about fifteen years older than Wundt himself, though by 1860 it was clear that their work was transforming the discipline of physiology. What the young Wundt obviously hoped was that this recent rather impressive success might serve as a model for the transformation of another field, namely psychology.²²

Although his immediate predecessors in German physiology provided Wundt with his most patent source of inspiration, it was not this group that had invented the project of an experimental science of physiology. They were in fact following in the footsteps of French investigators earlier in the century, chief among them François Magendie (1783–1855). In post-Napoleonic Paris we find the first clearly defined and established community of investigators devoted to the pursuit of physiology as an experimental science. Although physiological experiments had been performed before, and others engaged in this practice, more than merely a difference of degree existed between those who sometimes resorted to forms of experimentation, embedded in other practices, and those for whom experimentation had become the *via regia* to physiological knowledge.²³ The elevation of experimentation to the defining practice of scientific physiology cannot be separated from fundamental changes in the conception of the field and of its characteristic objects of investigation.

Traditionally, physiology had not been considered a separate discipline but merely the junior partner of anatomy. Medical schools had common chairs of anatomy and physiology, with the latter usually understood as being in the subordinate position.²⁴ Associated with this institutional arrangement was a certain conception of the subject matter of physiology. Function was considered to be subordinate to structure; one started with the anatomical organ and looked for its specific function.²⁵ The body was a static hierarchy of organs, each with its characteristic function. Thus, questions of physiological function could only arise after the structures to which the functions belonged had been established anatomically. The unit of investigation was the visible anatomical element, and the preferred method was that of dissection.

The situation changed drastically with a development that was as much a change in the object of investigation as a change of method. Functions were no longer seen as the properties of visible anatomical units but as abstract objects of investigation that might involve several organs as well as invisible processes. Thus it was no longer the function of a specific organ, like the stomach, which was at issue, but the role of the stomach in a

function like nutrition. Structures were now subordinated to general functions that involved the interplay of many organs and systems. A functional perspective was becoming dominant, and the most appropriate method for investigating the functions of living systems was that of experimental intervention.²⁶ The systematic use of experimentation ensured the predominance of the functional perspective, for the only kinds of questions that were asked and answered within this investigative framework were functional questions. Experimentation meant formulating one's inquiry in terms of the contribution of various factors to a particular functional effect. The choice of one's primary method entailed a certain conception of the object of investigation. For the science of physiology, experimental laboratory methods provided the "cognitive identity" that was needed to force its institutional separation from anatomy and to establish it firmly as an independent discipline, a development that was not finalized until well into the nineteenth century.²⁷

Not only did the transformation of physiology extend forward beyond the first generation of systematic experimentalists; its roots can also be traced backward to an earlier generation. This becomes particularly clear in those aspects of physiological thought that impinge most closely on psychology. After the middle of the eighteenth century one can detect a reaction against an earlier tradition that had made an absolute separation between mental and physical causes and effects, between voluntary (mentally caused) and involuntary (physically caused) action. Fundamental issues had been decided on the basis of whether one was thought to be dealing with a metaphysical mind substance or a metaphysical body substance. But in the second half of the eighteenth century a number of medical investigators – notably Whytt, Unzer, Prochaska, and, to some extent, Haller – began to move away from this preoccupation and to formulate questions of animal motion in purely functional terms.²⁸ The antecedents of the actions of living beings were now defined in terms of their effects rather than in terms of their status as mental or physical entities, and with this we get the emergence of the modern conception of *stimulation*. One result of this was to legitimize the treatment of psychological topics in a physiological context.²⁹ From then on the subject matter of psychology was likely to be influenced as much by developments in physiology as by philosophical considerations.

In the long run, there were two major channels through which an experimental physiology of function influenced investigative practices in psychology. One of these involved the functional characteristics of reflex movement. Although this aspect had some early effects on psychological theorizing, it did not influence psychological practice until well into the twentieth century, and so will not be considered here.³⁰ A second channel involving sensory physiology became active at a much earlier date. When the new program of systematic experimentation became extended to sen-

sory functions, some psychological implications could not be avoided. In these physiological experiments variations in sensory effects were investigated as a function of variations in the conditions of stimulation. But in terms of the prevailing metaphysic, sensory effects, at least in humans, were categorized as belonging to that private world of individual experience that was supposed to constitute the subject matter of psychology or mental philosophy. Thus, experiments in sensory physiology acquired a theoretical significance that often went considerably beyond the modest aims of their originators.³¹

When Wundt published the first textbook of the new experimental psychology, a good two-thirds of it consisted of an account of the physiology of the nervous system and of research in sensory physiology.³² The effect of the latter was to put the discussion of psychological objects – that is, sensory experience – in a functional context. This was a necessary consequence of the decision to ground psychological discourse in the experimental practices characteristic of sensory physiology. The kinds of questions that could be addressed on this basis were questions about the functional dependence of aspects of sensory experience on conditions of stimulation, such as intensity, spatial location, and temporal duration. This was the form taken by most experimental psychological investigations³³ and also the major empirical content of texts in the new discipline for many years to come. If the tradition of mental philosophy, with its notion of introspection as a method, bequeathed to the new psychology the concept of an inner mental world as a potential object of study, the model of physiological experimentation left the new discipline no choice but to pursue this study in a functional framework. The investigation of sensation and perception was virtually the only area in which these two approaches could be effectively combined, and the greater the pressure to expand beyond these confines the greater the methodological difficulties that the new discipline faced. We will explore some of these difficulties in the next chapter, but first we will need to take a closer look at the social features of early psychological experimentation.

The social practice of psychological experimentation

So far we have treated experimentation as though it were an almost purely cognitive activity. This perspective, however, ignores the crucial point about modern scientific experimentation – what distinguishes it from alchemical experimentation, for example – and that is its public character. We are not dealing with a private activity, whose procedures are to be kept secret, but with a technique for producing a social consensus about “the facts.” Scientific experiments are supposed to deal in phenomena that are in principle accessible to everyone and in procedures that should be replicable by all others with the necessary training and materials. A suc-

cessful program of scientific experiments requires a *community* of investigators who are able to agree on the veracity of certain observations, either because they have actually shared in their replication, or because they are satisfied that they have sufficient information to produce the same conjunction of events if they wanted to take the trouble to do so. The effective functioning of such a community requires the acceptance of certain rules and conventions about the conditions under which phenomena are produced and witnessed, and about the manner in which such matters are to be publicly communicated. Experimentation is not just a matter of cognitive construction but also a matter of social arrangement.

The original modern experimental community was that which formed around the Royal Society in England just after the middle of the seventeenth century.³⁴ Subsequently, new communities of the same general type formed in other centers and by the nineteenth century such communities generally had a more specialized focus, like experimental physiology. With each extension of the basic model some specific variants of the rules and conventions governing the life of the scientific community had to be adopted because of local conditions or special features of the objects of investigation. But this did not affect the fundamental features that characterized all such communities. They remained devoted to the engineering of an internal social consensus on so-called matters of fact by means of certain techniques that were common to them all. These techniques converged on the production of a sharp separation between matters of factual observation, about which all would necessarily have to agree, and matters of theoretical interpretation, about which differences of opinion within the community were permissible. Among the techniques deployed by the experimental community, those that regulate public communications within the community and the social relations of members of the community are of particular interest here.

The critical step that Wundt took toward the formation of a new experimental discipline was not the publication of his textbook, significant though this was, but the setting aside of a special space, identified as a psychological laboratory, for the regular conduct of psychological experiments by his senior students and himself. This happened at the University of Leipzig about five years after the first appearance of his textbook, and it was soon followed by further steps toward the institutionalization of this new area of experimental practice.³⁵ In 1883 the Leipzig laboratory received official recognition and a budget as a scientific institute, and in the same year there appeared the first number of a new journal, *Philosophische Studien*, which regularly published experimental reports from this laboratory. Wundt was fortunate in finding himself in the right place at the right time. The University of Leipzig was second only to Berlin in size and endowment at a time when the whole German university system was expanding rapidly,³⁶ and the prestige of German science and of the experi-

mental method were at their height. He quickly gathered in his laboratory a respectable number of students, many of whom proceeded to produce doctoral dissertations based on their experimental work.³⁷

Rather than work in isolation, several students would be doing experimental work at any one time, sharing facilities and assisting each other in various ways. They saw themselves as working in the same field, they shared certain theoretical interests, and their various experimental projects frequently had a bearing on each other. In other words, these students, together with Wundt himself, formed a viable community of experimenters. As some graduated and left, others came to take their place, so that practices that had become traditional in the community not only continued within it but were transplanted to other centers where the newly qualified experimental psychologists sometimes attempted to establish laboratories on the Leipzig model.³⁸ In Germany an independent second laboratory emerged at the University of Göttingen, and in 1890 a second journal, the *Zeitschrift für Psychologie und Physiologie der Sinnesorgane*, made its appearance.

For this experimental community to thrive, as it obviously did, and to pursue its work with some sense of effectiveness, certain technologies were needed to regulate its internal life.³⁹ The material technology could be taken over virtually intact from experimental physiology. The literary conventions of communicating the procedures and results of experimental investigations could also be adopted from existing models, and it was only gradually that the special features of psychological investigation led to new devices in the textual presentation of experimental results. In the early years of the experimental psychological community, the regulation of the social relationships inherent in the practice of psychological research presented the most interesting problems.

The classical work in sensory physiology – that of E. H. Weber, Hering, Helmholtz, and, for that matter, Fechner – had essentially been carried out by single investigators with, at most, occasional assistance. They communicated their findings to their scientific community in monographs and papers of a form that had become fairly standard in Germany by the mid-nineteenth century. The question of communication only arose for them when their work came to be published. But Wundt, as we have noted, established his laboratory explicitly for the benefit of his students, so that they might have a place to do the experimental work, which, for many of them, would form the basis of their doctoral dissertations and their first scientific publications. In doing this he was simply following the general trend in Germany of translating the ideal linkage between teaching and research into specific organizational forms.⁴⁰ Increasingly, the new sciences were linking the production of knowledge to the training of their recruits at universities, institutes, or laboratories, and Wundt did not hesitate to adopt this pattern.⁴¹ But this meant that the work coming out of his lab-

oratory was essentially collaborative work in which several individuals worked on different aspects of the same problem or on related problems.⁴²

This organizational aspect of the research work in the Leipzig laboratory made it possible to share the tedium that characterized many of the investigations, especially in psychophysics. It also made it easier to exploit the possibilities of the relatively more complex recording apparatus that was becoming available.⁴³ But such advantages were available only insofar as the collaboration of research workers was extended from the usual convergence *between* investigations to a systematic division of labor *within* each investigation.

This was an extremely important development with the most profound implications for the nature of psychological research. The division of labor that was spontaneously adopted in Wundt's laboratory was none other than the well-known division between the roles of "experimenter" and "subject" in psychological experiments. This division had no fundamental theoretical significance for the Leipzig group and was undertaken as an essentially practical response to the conditions of psychological research in Wundt's laboratory. As in other scientific experiments the observational data that constituted the experimental yield consisted mostly of readings of instruments (e.g., time measurements) and simple sensory judgments (e.g., on the relative brightness or size of two presented physical targets). But these targets were categorized as psychological stimuli, that is, as events whose status depended on an individual's conscious response. In other words, variations in the sensory target were of interest not because of what they might indicate about its own physical nature but because of what they might signify about the nature of the consciousness that responded to them.

But this change of the interpretive framework within which individual observations were placed entailed certain changes in the way in which individual investigators interacted with the apparatus and with other investigators who might be assisting with the experiment. The individual consciousness, being the object of investigation, had to be shielded from variable internal and external influences of unknown effect, which might distort the particular response that was of interest. So it seemed desirable to get *immediate* responses that allowed no time for reflection and to keep the responding individual in ignorance of the precise short-term variations in the stimulus conditions to which he was to respond. (This is not to be confused with being ignorant of the overall purpose of the experiment, which was definitely not thought desirable.)⁴⁴ But this made it increasingly difficult for individuals to experiment on themselves without assistance. The task of simultaneously manipulating the apparatus and playing the role of the possessor of a shielded private consciousness whose precise responses were the object of investigation was not easy, and was sometimes downright impossible. To share the burden, one made use of the availability of other members of the laboratory, or of obliging friends.⁴⁵ However, the

interpretive framework within which the work was done ensured that this collaboration would take the form of a division of labor between those who manipulated the apparatus and those whose consciousness was under investigation.

The effect of these practical measures was certainly not foreseen and hardly noticed for several generations. What occurred in practice was the development of a fundamental difference between the social conditions of experimentation in the natural sciences and in psychology. In the natural sciences any division of labor within an experimental investigation was unconnected with the fundamental relationship of the investigator to the object of investigation. One person might take charge of the distillation process while another might handle the weighing of the residue, but this did not affect their relation to the chemical object in any fundamental way. However, in psychological experiments one person would function as the repository of the object of investigation, of the data source, while the other would merely act as experimental manipulator in the usual way. This meant that whenever this division of labor was adopted the outcome of the investigation was the product of a social interaction within a role system whose structure was intimately connected with the way in which the object of investigation had been defined. In the natural sciences any division of labor in the experimental situation could continue more or less on an ad hoc basis, but in psychological experimentation the division of labor between experimenters and experimental subjects rather quickly developed into a universally accepted structural feature of the psychological experiment as such.

Once the organized psychological laboratories were established, self-experimentation began to decline in importance until, quite soon, it contributed only a small fraction of published experimental reports. This appears to have been the result of the conjunction of two factors. One factor must be looked for in the general development of institutional forms for scientific work, which favored collaborative research in organized laboratory units. This was not peculiar to psychology, except that psychology was affected at the very beginning of its existence as an experimental science, so that it hardly knew any experimental traditions other than highly organized ones. The second factor was connected with psychology's peculiar subject matter, which was the individual private consciousness. This made it inevitable that whenever an experimental division of labor was adopted it would take the form of a division between the roles of experimental manipulator and experimental subject or data source. As a result, the product of psychological experimentation depended not only on the interaction of investigators who all had the same fundamental relation to the object of investigation, but also depended on a unique kind of interaction between the experimenter's role and the experimental subject's role.

The special social organization of the psychological experiment is re-

flected in certain linguistic terms that appear even in the earliest published reports of these experiments. When describing procedural aspects of experiments, the research articles had to refer not only to the apparatus but also to the human participants in the experimental situation. Generally, these human participants were referred to in terms of the activity they had been assigned by the experimental division of labor. So we get references to “the discriminator,” “the associator,” or “the reactor,” depending on the particular activity that the experimental task required. Similarly, those functioning as what we would now call experimenters were referred to variously as “the manipulator,” “the signaler,” and “the reader” (*der Ablesende*), because of their function of manipulating the apparatus, signaling the experimental stimuli, or reading off the experimental results.⁴⁶ While the participants were also identified by name elsewhere in the publication – a point to which we will return in a later chapter – the procedural references in these early research articles clearly describe an interaction system, not among historical individuals, but among the occupants of specific roles within an experimental division of labor.

If one traces the way in which participants in psychological experiments are referred to in the early empirical papers of the Leipzig group, one can detect evidence of a rapid process of institutionalization, which manifests itself in two ways. First, the more concrete terms used to refer to experimental participants tend to drop out and to be replaced by more abstract terms. The relatively specific activities that the experimental division of labor had assigned to them – activities like signaling or associating, as already noted⁴⁷ – were soon replaced by labels that refer in a more general way to the function of the participant in psychological investigation. The favored terms that now emerge are “observer,” “reactor” (*Reagent*), and “person under experiment” (*Versuchsperson*) on the one hand and “experimenter” on the other. This means that individuals performing very different specific activities in different experiments are now referred to by the same general term. In other words, linguistic usage begins to reflect the fact that all psychological experiments involve a system of distributing certain general functions among the participants, and that these general functions are not tied to any specific activity.

The second, related development of linguistic usage in the early experimental psychological literature involves a certain standardization of terminology. During the early years of the Leipzig laboratory there is no uniformity in the labels applied to the experimental participants. Even when two investigators are concerned with the same research topic, they might use different terms to refer to their experimental subjects. For instance, two Leipzig experimenters investigated the “time sense” within a couple of years of each other, yet the one refers to his subjects as “reactors,” whereas the other uses the term “observers.”⁴⁸ This lack of semantic uniformity is also reflected in the fact that some investigators use two or

three terms interchangeably within the same experimental report to refer to their experimental subjects.⁴⁹ But gradually usage becomes more standardized, even though complete standardization was not achieved till much later, as we will see in chapter 6. Before the end of the nineteenth century, however, most investigators settle into the consistent use of either of two terms to refer to the human data source in psychological experiments, the terms being “subject” (*Versuchsperson*) and “observer” (*Beobachter*).

The social process reflected in these developments clearly involved the institutionalization of psychological experimentation. In the 1880s the Leipzig laboratory evolved a distinct pattern of practice that defined for its participants what a psychological experiment looked like. Many of those who spent time at Wundt’s laboratory subsequently used its pattern of experimental practice as a model to be emulated when they attempted to establish their own laboratories elsewhere. This model character of the Leipzig laboratory extended not only to the inventory of apparatus⁵⁰ and the kinds of research problems that defined a psychological laboratory but also involved the social arrangements required by psychological experimentation. Thus there grew up a certain tradition of how the social organization of work in psychological laboratories was to be structured. It was a tradition that depended on a certain definition of the object of psychological investigation, on a specific heritage of experimentation in sensory physiology, and on some local historical conditions that gave a particular form to the social conditions of psychological experimentation. But the effectiveness of this tradition would extend only as far as the historical conjunction of these three factors. Changes in any of these factors would produce a different model of psychological experimentation. This development will be examined in the next two chapters.

3

Divergence of investigative practice: The repudiation of Wundt

Wundt's limitations on introspection, experiment, and scientific psychology

Wundt's historical role was in some ways akin to the fate of the sorcerer's apprentice. He successfully mobilized some very effective practical techniques in the service of certain limited goals, but then found that these techniques turned into forces that had passed completely out of his control and were about to destroy the very framework within which he had put them to work. This was true of the link between introspection and experiment, which he advocated, and of the form of experimental psychology that he put into practice. Wundt, as has been observed, had many students but no genuine disciples. If he was the father of a discipline, he was a father whose fate is more reminiscent of the myth of *Totem and Taboo* than the myth of Chronos. Virtually everything that happened in modern psychology was a repudiation of Wundt, explicitly or implicitly. To understand this process, we must try to understand its starting point, namely, Wundt's conception of psychological practice.

Wundt's theory and practice of *introspection* diverged quite sharply from that of many of his students. On this issue (as on several others) he never emerged from the shadow of Kant, which meant that he basically accepted the object of knowledge to which the method of introspection corresponded but denigrated the method itself. He never doubted that the private consciousness was the object that psychology had to study, but he agreed with Kant and later critics like Comte and Lange that introspection was not the instrument that would transform this object into a scientific object. In fact, he went so far as to ridicule the introspectionist, likening him to a Baron Münchhausen attempting to pull himself out of the bog by his own pigtail.¹ Yet Wundt thought he had found a way to make consciousness available